

Object Oriented Programming with Java 1 (2026)

Assignment 2

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1 Problem description

This assignment is concerned with the implementation and design of a program for analyzing text. Apart from prompting the user for a proper text file, the overall functionality of the program is to read the file, analyze it, and print a report of the text to the console. The report should consist of the following statistics

- Number of paragraphs in the text
- Sentences per paragraph
- Number of sentences in the text
- Words per sentence
- Number of words in the text
- Average word length
- Lix score of the text

The Lix score can be computed as the sum of the percentage of long words and the average number of words per sentence, i.e.

$$\text{Lix score} := \frac{A}{B} + 100 \frac{C}{A} \quad (1)$$

With A being the number of words, B being the number of sentences, and C the number of long words in a given text. You can define long words as words consisting of more than six characters.

2 Program structure and classes

The program entry point should be located in the file `Assignment2.java`. Segmenting a program into a collection of classes with limited responsibilities is one of the first steps to create more readable and extensible code, so in order to solve this assignment, you need to implement the following classes

- `TextReader`
- `TextAnalyzer`

- ParagraphAnalyzer
- SentenceAnalyzer
- WordAnalyzer

The naming of the classes mentioned should be indicative of their responsibilities within the greater program, but is here given a few words.

The TextReader class is responsible for reading the text that is to be analyzed, and it should thus have a method with the following signature

```
ReadFile : () -> String
```

which prompts the user for a path to a text file, which is read and stored in a String object before being returned.

The TextAnalyzer class is the class having the broadest coupling to the other parts of the program. TextAnalyzer should in some way be able to process the String obtained by the TextReader class. Furthermore it should have a 'has-a' relationship with the other analyzer classes.

ParagraphAnalyzer is responsible for analyzing the text on the level of paragraphs, and should therefore be able to split the given text into a collection of paragraphs. The SentenceAnalyzer is responsible for the sentence level of analysis, and in order to do this, the class should be able to turn the paragraphs into a collection of sentences. Finally, the WordAnalyzer analyzes words, and should therefore be able to transform a collection of sentences into a collection of words. Notice how the analyzer classes depend on input from the analyzer one level up in the structure of text. Think carefully about how you want to implement this dependency and how data should be shared between classes.

Each of the analyzer classes should have a method analyze() which starts the process of analysis in order to obtain the relevant statistics.

Your program should be tested on the two text files theuglyduckling.txt and dengrimmeaelling.txt. These files should be in the same directory as your other java files in your submission.

Hint: The split() method of the String class can be valuable, as well as the Array or ArrayList data structure.

3 Submission

The assignment should be handed in on your GitHub repository in a folder named Assignment 2. Be sure to commit and push your solution.

The Moodle Assignment 2 submission is a text file named Assignment2.txt which only contains the URL for your GitHub repository.

The assignment should be handed in no later than 23:59 on April 24.